Lateralus Language Specification v3

Normative Reference for Lateralus v3.0 -- Syntax, Semantics, Standard Library

Lateralus Language

bad-antics · April 2026 · Lateralus Language Research

ABSTRACT Lateralus Language -- lateralus.dev

1. Language Overview and Goals

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Details

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

```
// Module structure
module geometry

import math

pub struct Point {
    x: float,
    y: float,
}

pub fn distance(a: Point, b: Point) -> float {
    let dx = b.x - a.x
    let dy = b.y - a.y
    return math.sqrt(dx*dx + dy*dy)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

2. Notation Used in This Specification

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

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Details

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no

other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

```
// Enum with data variants
pub enum Shape {
    Circle { center: Point, radius: float },
    Rect   { origin: Point, width: float, height: float },
    Triangle { a: Point, b: Point, c: Point },
}

pub fn area(s: Shape) -> float {
    match s {
        Shape.Circle { radius: r, .. }       => math.PI * r * r,
        Shape.Rect   { width: w, height: h, .. } => w * h,
        Shape.Triangle { a, b, c }           => triangle_area(a, b, c),
    }
}
```

Keyword Kind Since Purpose

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measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

3. Lexical Structure: Characters and Encoding

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

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Details

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

```
// Generic function with trait bound
trait Summable {
    fn zero() -> Self
    fn add(self, other: Self) -> Self
}

fn sum_all[T: Summable](items: list[T]) -> T {
    return items |> reduce(T.zero(), fn(acc, x) { acc.add(x) })
}

impl Summable for int {
    fn zero() -> int { return 0 }
    fn add(self, other: int) -> int { return self + other }
}
```

Keyword Kind Since Purpose

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4. Lexical Structure: Whitespace and Comments

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

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Details

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

```
// Pattern matching -- exhaustiveness
fn describe_result(r: Result[int, str]) -> str {
    match r {
        Ok(n) if n > 100 => "large positive",
        Ok(n) if n > 0   => "small positive",
        Ok(0)            => "zero",
```

```
        Ok(n)            => "negative: " + str(n),
        Err(e)           => "error: " + e,
        // All cases covered -- no warning
    }

}
```

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measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

5. Lexical Structure: Identifiers

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait

by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Details

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

```
// Pipeline with types at each stage
fn pipeline_example(nums: list[int]) -> map {
    let analysis = nums
        |> filter(fn(x) { x > 0 })                    // list[int]
        |> map(fn(x: int) -> float { sqrt(x as float) }) // list[float]
        |> take(10)                                     // list[float]

        |> fold({ "sum": 0.0, "count": 0 }, fn(acc, x) {
            acc["sum"]   = acc["sum"] + x
            acc["count"] = acc["count"] + 1
            return acc
        })  // map
    return analysis
}
```

Keyword Kind Since Purpose

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let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

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6. Lexical Structure: Keywords

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Details

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

```
// Async function and await

async fn fetch_and_parse(url: str) -> Result[Data, str] {
    let response = await http.get(url)
    let text     = await response.text()
    match parse_json(text) {
        Ok(json) => Ok(Data.from_json(json)),
        Err(e)   => Err("Parse error: " + e),
    }
}

async fn main() {
    let tasks = URLS |> map(fn(url) { fetch_and_parse(url) })
    let results = await gather(tasks)   // concurrent execution
    results |> filter(fn(r) { r.is_ok() }) |> for_each(println)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

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match expression v1.0 pattern matching

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async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

7. Lexical Structure: Literals: Integers

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

Details

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace,

starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task

scheduler.

```
// Ownership and borrowing
fn process(data: list[int]) -> int {
    let total = sum(data)     // borrows data (immutable)
    let mean  = total / len(data)  // data still accessible
    return mean
}
// data is NOT moved -- process only borrows it
```

```
fn move_example() {
    let v = [1, 2, 3]
    let w = v               // v is MOVED into w
    // println(v)           // ERROR: v was moved
    println(w)              // OK
}
```

Keyword Kind Since Purpose

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let statement v1.0 variable binding

mut modifier v1.0 mutable variable

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async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

8. Lexical Structure: Literals: Floats

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses

monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

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module provides async/await and the task scheduler.

Details

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

```
// Module structure
module geometry

import math

pub struct Point {
    x: float,
    y: float,
}

pub fn distance(a: Point, b: Point) -> float {
    let dx = b.x - a.x
    let dy = b.y - a.y
    return math.sqrt(dx*dx + dy*dy)
}
```

Keyword Kind Since Purpose

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measure statement v1.4 performance profiling

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from...import statement v1.5 selective import

9. Lexical Structure: Literals: Strings

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

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Details

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

```
// Enum with data variants
pub enum Shape {
    Circle { center: Point, radius: float },
    Rect   { origin: Point, width: float, height: float },
    Triangle { a: Point, b: Point, c: Point },
}

pub fn area(s: Shape) -> float {
    match s {
        Shape.Circle { radius: r, .. }       => math.PI * r * r,
        Shape.Rect   { width: w, height: h, .. } => w * h,
        Shape.Triangle { a, b, c }           => triangle_area(a, b, c),
```

```
    }
}
```

Keyword Kind Since Purpose

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let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

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from...import statement v1.5 selective import

10. Lexical Structure: Literals: Booleans and Nil

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the

compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through

interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

Details

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

```
// Generic function with trait bound
trait Summable {
    fn zero() -> Self
    fn add(self, other: Self) -> Self
}

fn sum_all[T: Summable](items: list[T]) -> T {
    return items |> reduce(T.zero(), fn(acc, x) { acc.add(x) })
}

impl Summable for int {
    fn zero() -> int { return 0 }
    fn add(self, other: int) -> int { return self + other }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

11. Lexical Structure: Operators and Punctuation

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM

supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

Details

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

```
// Pattern matching -- exhaustiveness
fn describe_result(r: Result[int, str]) -> str {
    match r {
        Ok(n) if n > 100 => "large positive",
        Ok(n) if n > 0   => "small positive",
        Ok(0)            => "zero",
        Ok(n)            => "negative: " + str(n),
        Err(e)           => "error: " + e,
        // All cases covered -- no warning
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

12. Grammar: Top-Level Structure (EBNF)

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The

'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Details

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

```
// Pipeline with types at each stage
fn pipeline_example(nums: list[int]) -> map {
    let analysis = nums
```

```
        |> filter(fn(x) { x > 0 })                    // list[int]
        |> map(fn(x: int) -> float { sqrt(x as float) }) // list[float]
        |> take(10)                                     // list[float]
        |> fold({ "sum": 0.0, "count": 0 }, fn(acc, x) {
            acc["sum"]   = acc["sum"] + x
            acc["count"] = acc["count"] + 1
            return acc
        })  // map
    return analysis
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

13. Grammar: Declarations

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Details

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

```
// Async function and await
async fn fetch_and_parse(url: str) -> Result[Data, str] {
    let response = await http.get(url)
    let text     = await response.text()
    match parse_json(text) {
        Ok(json) => Ok(Data.from_json(json)),
        Err(e)   => Err("Parse error: " + e),
    }
}

async fn main() {
    let tasks = URLS |> map(fn(url) { fetch_and_parse(url) })
    let results = await gather(tasks)   // concurrent execution
    results |> filter(fn(r) { r.is_ok() }) |> for_each(println)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

14. Grammar: Statements

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

Details

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

```
// Ownership and borrowing
fn process(data: list[int]) -> int {
    let total = sum(data)     // borrows data (immutable)
    let mean  = total / len(data)  // data still accessible
    return mean
}
// data is NOT moved -- process only borrows it

fn move_example() {
    let v = [1, 2, 3]
    let w = v               // v is MOVED into w
    // println(v)           // ERROR: v was moved
    println(w)              // OK
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

15. Grammar: Expressions

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Details

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e),

Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

```
// Module structure
module geometry

import math

pub struct Point {
    x: float,
    y: float,
}

pub fn distance(a: Point, b: Point) -> float {
    let dx = b.x - a.x
    let dy = b.y - a.y
    return math.sqrt(dx*dx + dy*dy)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

16. Grammar: Patterns

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Details

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

```
// Enum with data variants
pub enum Shape {
    Circle { center: Point, radius: float },
    Rect   { origin: Point, width: float, height: float },
    Triangle { a: Point, b: Point, c: Point },
}

pub fn area(s: Shape) -> float {
    match s {
        Shape.Circle { radius: r, .. }       => math.PI * r * r,
        Shape.Rect   { width: w, height: h, .. } => w * h,
        Shape.Triangle { a, b, c }           => triangle_area(a, b, c),

    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

17. Grammar: Types

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Details

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness

checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

```
// Generic function with trait bound
trait Summable {
    fn zero() -> Self
    fn add(self, other: Self) -> Self
}

fn sum_all[T: Summable](items: list[T]) -> T {
    return items |> reduce(T.zero(), fn(acc, x) { acc.add(x) })
}

impl Summable for int {
    fn zero() -> int { return 0 }
    fn add(self, other: int) -> int { return self + other }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

18. Grammar: Operator Precedence Table

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Details

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

```
// Pattern matching -- exhaustiveness
fn describe_result(r: Result[int, str]) -> str {
    match r {
        Ok(n) if n > 100 => "large positive",
        Ok(n) if n > 0   => "small positive",
        Ok(0)            => "zero",
        Ok(n)            => "negative: " + str(n),
        Err(e)           => "error: " + e,
        // All cases covered -- no warning
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

19. Semantics: Values and Types

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

Details

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The

'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

```
// Pipeline with types at each stage
fn pipeline_example(nums: list[int]) -> map {
    let analysis = nums
        |> filter(fn(x) { x > 0 })                    // list[int]
        |> map(fn(x: int) -> float { sqrt(x as float) }) // list[float]
        |> take(10)                                     // list[float]
        |> fold({ "sum": 0.0, "count": 0 }, fn(acc, x) {
            acc["sum"]   = acc["sum"] + x
            acc["count"] = acc["count"] + 1
            return acc
        })  // map
    return analysis
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

20. Semantics: Variables and Bindings

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

Details

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

```
// Async function and await
async fn fetch_and_parse(url: str) -> Result[Data, str] {
    let response = await http.get(url)
    let text     = await response.text()
    match parse_json(text) {
        Ok(json) => Ok(Data.from_json(json)),
        Err(e)   => Err("Parse error: " + e),
    }
}

async fn main() {
    let tasks = URLS |> map(fn(url) { fetch_and_parse(url) })
    let results = await gather(tasks)   // concurrent execution
    results |> filter(fn(r) { r.is_ok() }) |> for_each(println)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

21. Semantics: Mutation (let mut)

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM

supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

Details

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

```
// Ownership and borrowing
fn process(data: list[int]) -> int {
    let total = sum(data)     // borrows data (immutable)
    let mean  = total / len(data)  // data still accessible
    return mean
}
// data is NOT moved -- process only borrows it

fn move_example() {
    let v = [1, 2, 3]
    let w = v               // v is MOVED into w
    // println(v)           // ERROR: v was moved
    println(w)              // OK
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

22. Semantics: Scope and Shadowing

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime.

The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

Details

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system

should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

```
// Module structure
module geometry

import math

pub struct Point {
    x: float,
    y: float,
}

pub fn distance(a: Point, b: Point) -> float {
    let dx = b.x - a.x
    let dy = b.y - a.y
    return math.sqrt(dx*dx + dy*dy)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

23. Semantics: Closures

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation

to be executed when the awaited value is ready.

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

Details

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

```
// Enum with data variants
pub enum Shape {
    Circle { center: Point, radius: float },
    Rect   { origin: Point, width: float, height: float },
    Triangle { a: Point, b: Point, c: Point },
}

pub fn area(s: Shape) -> float {
    match s {
        Shape.Circle { radius: r, .. }       => math.PI * r * r,
        Shape.Rect   { width: w, height: h, .. } => w * h,
        Shape.Triangle { a, b, c }           => triangle_area(a, b, c),
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

24. Semantics: Function Calls and Arguments

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A

module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Details

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

```
// Generic function with trait bound
trait Summable {
    fn zero() -> Self
    fn add(self, other: Self) -> Self
}

fn sum_all[T: Summable](items: list[T]) -> T {
    return items |> reduce(T.zero(), fn(acc, x) { acc.add(x) })
```

```
}

impl Summable for int {
    fn zero() -> int { return 0 }
    fn add(self, other: int) -> int { return self + other }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

25. Semantics: Control Flow: if/else

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as

Type). Function calls, match expressions, and blocks are all expressions that produce values.

Details

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

```
// Pattern matching -- exhaustiveness
fn describe_result(r: Result[int, str]) -> str {
    match r {
        Ok(n) if n > 100 => "large positive",
        Ok(n) if n > 0   => "small positive",
        Ok(0)            => "zero",
        Ok(n)            => "negative: " + str(n),
        Err(e)           => "error: " + e,
        // All cases covered -- no warning
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

26. Semantics: Control Flow: while

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

Details

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

```
// Pipeline with types at each stage
fn pipeline_example(nums: list[int]) -> map {
    let analysis = nums
        |> filter(fn(x) { x > 0 })                    // list[int]
        |> map(fn(x: int) -> float { sqrt(x as float) }) // list[float]
        |> take(10)                                     // list[float]
        |> fold({ "sum": 0.0, "count": 0 }, fn(acc, x) {
            acc["sum"]   = acc["sum"] + x
            acc["count"] = acc["count"] + 1
            return acc
        })  // map
    return analysis
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

27. Semantics: Control Flow: for-in

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Details

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

```
// Async function and await
async fn fetch_and_parse(url: str) -> Result[Data, str] {
    let response = await http.get(url)
    let text     = await response.text()
    match parse_json(text) {
        Ok(json) => Ok(Data.from_json(json)),
        Err(e)   => Err("Parse error: " + e),
    }
}

async fn main() {
    let tasks = URLS |> map(fn(url) { fetch_and_parse(url) })
    let results = await gather(tasks)   // concurrent execution
    results |> filter(fn(r) { r.is_ok() }) |> for_each(println)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

28. Semantics: Control Flow: match

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope,

the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Details

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

```
// Ownership and borrowing
fn process(data: list[int]) -> int {
    let total = sum(data)     // borrows data (immutable)
    let mean  = total / len(data)  // data still accessible
    return mean
}
// data is NOT moved -- process only borrows it

fn move_example() {
    let v = [1, 2, 3]
    let w = v               // v is MOVED into w
    // println(v)           // ERROR: v was moved
    println(w)              // OK
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

29. Semantics: Control Flow: return/break/continue

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Details

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

```
// Module structure
module geometry

import math

pub struct Point {
    x: float,
    y: float,
}
```

```
pub fn distance(a: Point, b: Point) -> float {
    let dx = b.x - a.x
    let dy = b.y - a.y
    return math.sqrt(dx*dx + dy*dy)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

30. Type System: Primitive Types

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for

fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Details

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

```
// Enum with data variants
pub enum Shape {
    Circle { center: Point, radius: float },

    Rect   { origin: Point, width: float, height: float },
    Triangle { a: Point, b: Point, c: Point },
}

pub fn area(s: Shape) -> float {
    match s {
        Shape.Circle { radius: r, .. }       => math.PI * r * r,
        Shape.Rect   { width: w, height: h, .. } => w * h,
        Shape.Triangle { a, b, c }           => triangle_area(a, b, c),
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

31. Type System: Composite Types (struct, enum)

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

Details

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The

'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

```
// Generic function with trait bound
trait Summable {
    fn zero() -> Self
    fn add(self, other: Self) -> Self
}

fn sum_all[T: Summable](items: list[T]) -> T {
    return items |> reduce(T.zero(), fn(acc, x) { acc.add(x) })
}

impl Summable for int {
    fn zero() -> int { return 0 }
    fn add(self, other: int) -> int { return self + other }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

32. Type System: Generic Types

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

Details

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

```
// Pattern matching -- exhaustiveness
fn describe_result(r: Result[int, str]) -> str {
    match r {
        Ok(n) if n > 100 => "large positive",
        Ok(n) if n > 0   => "small positive",
        Ok(0)            => "zero",
        Ok(n)            => "negative: " + str(n),
        Err(e)           => "error: " + e,
        // All cases covered -- no warning
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

33. Type System: Trait Bounds

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

Details

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

```
// Pipeline with types at each stage
fn pipeline_example(nums: list[int]) -> map {
    let analysis = nums
        |> filter(fn(x) { x > 0 })                    // list[int]
        |> map(fn(x: int) -> float { sqrt(x as float) }) // list[float]
        |> take(10)                                     // list[float]
        |> fold({ "sum": 0.0, "count": 0 }, fn(acc, x) {
            acc["sum"]   = acc["sum"] + x
            acc["count"] = acc["count"] + 1
            return acc
        })  // map
    return analysis
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

34. Type System: Type Inference (Algorithm W)

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The

'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

Details

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

```
// Async function and await
async fn fetch_and_parse(url: str) -> Result[Data, str] {
    let response = await http.get(url)
    let text     = await response.text()
    match parse_json(text) {
        Ok(json) => Ok(Data.from_json(json)),
        Err(e)   => Err("Parse error: " + e),
    }
}

async fn main() {
    let tasks = URLS |> map(fn(url) { fetch_and_parse(url) })
    let results = await gather(tasks)   // concurrent execution
    results |> filter(fn(r) { r.is_ok() }) |> for_each(println)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

35. Type System: Unification

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the

module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

Details

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

```
// Ownership and borrowing
fn process(data: list[int]) -> int {
    let total = sum(data)     // borrows data (immutable)
    let mean  = total / len(data)  // data still accessible
    return mean
}
// data is NOT moved -- process only borrows it
```

```
fn move_example() {
    let v = [1, 2, 3]
    let w = v               // v is MOVED into w
    // println(v)           // ERROR: v was moved
    println(w)              // OK
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

36. Type System: Occurs Check

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Details

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

```
// Module structure
module geometry

import math

pub struct Point {
    x: float,
    y: float,
}

pub fn distance(a: Point, b: Point) -> float {
    let dx = b.x - a.x
    let dy = b.y - a.y
    return math.sqrt(dx*dx + dy*dy)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

37. Type System: Let Polymorphism

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles:

clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Details

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

```
// Enum with data variants
```

```
pub enum Shape {
    Circle { center: Point, radius: float },
    Rect   { origin: Point, width: float, height: float },
    Triangle { a: Point, b: Point, c: Point },
}

pub fn area(s: Shape) -> float {
    match s {
        Shape.Circle { radius: r, .. }       => math.PI * r * r,
        Shape.Rect   { width: w, height: h, .. } => w * h,
        Shape.Triangle { a, b, c }           => triangle_area(a, b, c),
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

38. Type System: Subtyping and Coercion

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as

Type). Function calls, match expressions, and blocks are all expressions that produce values.

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

Details

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

```
// Generic function with trait bound
trait Summable {
    fn zero() -> Self
    fn add(self, other: Self) -> Self
}

fn sum_all[T: Summable](items: list[T]) -> T {
    return items |> reduce(T.zero(), fn(acc, x) { acc.add(x) })
}

impl Summable for int {
    fn zero() -> int { return 0 }
    fn add(self, other: int) -> int { return self + other }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

39. Memory Model: Ownership

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Details

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

```
// Pattern matching -- exhaustiveness
fn describe_result(r: Result[int, str]) -> str {
    match r {
        Ok(n) if n > 100 => "large positive",
        Ok(n) if n > 0   => "small positive",
        Ok(0)            => "zero",
        Ok(n)            => "negative: " + str(n),
        Err(e)           => "error: " + e,
        // All cases covered -- no warning
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

40. Memory Model: Borrowing

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Details

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support

default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

```
// Pipeline with types at each stage
fn pipeline_example(nums: list[int]) -> map {
    let analysis = nums
        |> filter(fn(x) { x > 0 })                    // list[int]
        |> map(fn(x: int) -> float { sqrt(x as float) }) // list[float]
        |> take(10)                                     // list[float]
        |> fold({ "sum": 0.0, "count": 0 }, fn(acc, x) {
            acc["sum"]   = acc["sum"] + x
            acc["count"] = acc["count"] + 1
            return acc
        })  // map
    return analysis
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

41. Memory Model: Lifetimes

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Details

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

```
// Async function and await
async fn fetch_and_parse(url: str) -> Result[Data, str] {
    let response = await http.get(url)
    let text     = await response.text()
    match parse_json(text) {
        Ok(json) => Ok(Data.from_json(json)),
        Err(e)   => Err("Parse error: " + e),
    }
}

async fn main() {
    let tasks = URLS |> map(fn(url) { fetch_and_parse(url) })
    let results = await gather(tasks)   // concurrent execution
    results |> filter(fn(r) { r.is_ok() }) |> for_each(println)

}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

42. Memory Model: Move Semantics

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Details

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics)

at the cost of potentially larger binary sizes.

```
// Ownership and borrowing
fn process(data: list[int]) -> int {
    let total = sum(data)     // borrows data (immutable)
    let mean  = total / len(data)  // data still accessible
    return mean
}
// data is NOT moved -- process only borrows it
```

```
fn move_example() {
    let v = [1, 2, 3]
    let w = v               // v is MOVED into w
    // println(v)           // ERROR: v was moved
    println(w)              // OK
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

43. Memory Model: Copy Trait

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical

pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

Details

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

```
// Module structure

module geometry

import math

pub struct Point {
    x: float,
    y: float,
}

pub fn distance(a: Point, b: Point) -> float {
    let dx = b.x - a.x
    let dy = b.y - a.y
    return math.sqrt(dx*dx + dy*dy)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

44. Collections: list[T]

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

Details

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final

compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

```
// Enum with data variants
pub enum Shape {
    Circle { center: Point, radius: float },
    Rect   { origin: Point, width: float, height: float },
    Triangle { a: Point, b: Point, c: Point },
}

pub fn area(s: Shape) -> float {
    match s {
        Shape.Circle { radius: r, .. }       => math.PI * r * r,
        Shape.Rect   { width: w, height: h, .. } => w * h,
        Shape.Triangle { a, b, c }           => triangle_area(a, b, c),
```

```
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

45. Collections: map[K,V]

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation

to be executed when the awaited value is ready.

Details

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

```
// Generic function with trait bound
trait Summable {
    fn zero() -> Self
    fn add(self, other: Self) -> Self
}

fn sum_all[T: Summable](items: list[T]) -> T {
    return items |> reduce(T.zero(), fn(acc, x) { acc.add(x) })
}

impl Summable for int {
    fn zero() -> int { return 0 }
    fn add(self, other: int) -> int { return self + other }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

46. Collections: set[T]

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

Details

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

```
// Pattern matching -- exhaustiveness
fn describe_result(r: Result[int, str]) -> str {
    match r {
        Ok(n) if n > 100 => "large positive",
        Ok(n) if n > 0   => "small positive",
        Ok(0)            => "zero",
        Ok(n)            => "negative: " + str(n),
        Err(e)           => "error: " + e,
        // All cases covered -- no warning
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

47. String Operations

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports

circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

Details

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

```
// Pipeline with types at each stage
fn pipeline_example(nums: list[int]) -> map {
    let analysis = nums
        |> filter(fn(x) { x > 0 })                    // list[int]
        |> map(fn(x: int) -> float { sqrt(x as float) }) // list[float]
        |> take(10)                                     // list[float]
        |> fold({ "sum": 0.0, "count": 0 }, fn(acc, x) {
            acc["sum"]   = acc["sum"] + x
            acc["count"] = acc["count"] + 1
            return acc
        })  // map
    return analysis
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

48. Error Handling: Option[T]

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline

operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Details

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

```
// Async function and await
async fn fetch_and_parse(url: str) -> Result[Data, str] {
    let response = await http.get(url)
    let text     = await response.text()
    match parse_json(text) {
        Ok(json) => Ok(Data.from_json(json)),
        Err(e)   => Err("Parse error: " + e),
    }
}

async fn main() {
    let tasks = URLS |> map(fn(url) { fetch_and_parse(url) })
    let results = await gather(tasks)   // concurrent execution
    results |> filter(fn(r) { r.is_ok() }) |> for_each(println)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

49. Error Handling: Result[T,E]

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors),

and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Details

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

```
// Ownership and borrowing
fn process(data: list[int]) -> int {
    let total = sum(data)     // borrows data (immutable)
    let mean  = total / len(data)  // data still accessible
    return mean
}
// data is NOT moved -- process only borrows it

fn move_example() {
    let v = [1, 2, 3]
    let w = v               // v is MOVED into w
    // println(v)           // ERROR: v was moved
```

```
    println(w)              // OK
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

50. Error Handling: Propagation with ?

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

Details

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

```
// Module structure
module geometry

import math

pub struct Point {
    x: float,
    y: float,
}

pub fn distance(a: Point, b: Point) -> float {
    let dx = b.x - a.x
    let dy = b.y - a.y
    return math.sqrt(dx*dx + dy*dy)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

51. Pipeline Operator: Semantics

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Details

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

```
// Enum with data variants
pub enum Shape {
    Circle { center: Point, radius: float },
    Rect   { origin: Point, width: float, height: float },
    Triangle { a: Point, b: Point, c: Point },
}

pub fn area(s: Shape) -> float {
    match s {
        Shape.Circle { radius: r, .. }       => math.PI * r * r,
        Shape.Rect   { width: w, height: h, .. } => w * h,
        Shape.Triangle { a, b, c }           => triangle_area(a, b, c),
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

52. Pipeline Operator: Type Inference

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Details

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait

implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

```
// Generic function with trait bound
trait Summable {
    fn zero() -> Self
    fn add(self, other: Self) -> Self
}

fn sum_all[T: Summable](items: list[T]) -> T {
    return items |> reduce(T.zero(), fn(acc, x) { acc.add(x) })
}

impl Summable for int {
    fn zero() -> int { return 0 }
    fn add(self, other: int) -> int { return self + other }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

53. Pipeline Operator: Fusion Rules

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Details

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

```
// Pattern matching -- exhaustiveness
fn describe_result(r: Result[int, str]) -> str {
    match r {
        Ok(n) if n > 100 => "large positive",
        Ok(n) if n > 0   => "small positive",
        Ok(0)            => "zero",
        Ok(n)            => "negative: " + str(n),
        Err(e)           => "error: " + e,
        // All cases covered -- no warning
    }

}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

54. Pattern Matching: Exhaustiveness

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Details

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

```
// Pipeline with types at each stage
fn pipeline_example(nums: list[int]) -> map {
    let analysis = nums
        |> filter(fn(x) { x > 0 })                    // list[int]
        |> map(fn(x: int) -> float { sqrt(x as float) }) // list[float]
        |> take(10)                                     // list[float]
        |> fold({ "sum": 0.0, "count": 0 }, fn(acc, x) {

            acc["sum"]   = acc["sum"] + x
            acc["count"] = acc["count"] + 1
            return acc
        })  // map
    return analysis
```

```
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

55. Pattern Matching: Guard Clauses

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

Details

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

```
// Async function and await
async fn fetch_and_parse(url: str) -> Result[Data, str] {

    let response = await http.get(url)
    let text     = await response.text()
    match parse_json(text) {
        Ok(json) => Ok(Data.from_json(json)),
        Err(e)   => Err("Parse error: " + e),
    }
}

async fn main() {
    let tasks = URLS |> map(fn(url) { fetch_and_parse(url) })
    let results = await gather(tasks)   // concurrent execution
    results |> filter(fn(r) { r.is_ok() }) |> for_each(println)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

56. Traits: Declaration

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

Details

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final

compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

```
// Ownership and borrowing
fn process(data: list[int]) -> int {
    let total = sum(data)     // borrows data (immutable)
    let mean  = total / len(data)  // data still accessible
    return mean
}
// data is NOT moved -- process only borrows it

fn move_example() {
    let v = [1, 2, 3]
    let w = v               // v is MOVED into w
    // println(v)           // ERROR: v was moved
    println(w)              // OK
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

57. Traits: Implementation

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation

to be executed when the awaited value is ready.

Details

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles

by giving each module a declared interface before its implementation is compiled.

```
// Module structure
module geometry

import math

pub struct Point {
    x: float,
    y: float,
}

pub fn distance(a: Point, b: Point) -> float {
    let dx = b.x - a.x
    let dy = b.y - a.y
    return math.sqrt(dx*dx + dy*dy)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

58. Traits: Default Methods

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation

to be executed when the awaited value is ready.

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

Details

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

```
// Enum with data variants
pub enum Shape {
    Circle { center: Point, radius: float },
    Rect   { origin: Point, width: float, height: float },
    Triangle { a: Point, b: Point, c: Point },
}

pub fn area(s: Shape) -> float {
    match s {
        Shape.Circle { radius: r, .. }       => math.PI * r * r,
        Shape.Rect   { width: w, height: h, .. } => w * h,
        Shape.Triangle { a, b, c }           => triangle_area(a, b, c),
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

59. Traits: Standard Traits (Eq, Ord, Hash, Display)

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

Details

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

```
// Generic function with trait bound
trait Summable {
    fn zero() -> Self
    fn add(self, other: Self) -> Self
}

fn sum_all[T: Summable](items: list[T]) -> T {
    return items |> reduce(T.zero(), fn(acc, x) { acc.add(x) })
}
```

```
impl Summable for int {
    fn zero() -> int { return 0 }
    fn add(self, other: int) -> int { return self + other }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

60. Module System: Declarations and Imports

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the

module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

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definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Details

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

```
// Pattern matching -- exhaustiveness
fn describe_result(r: Result[int, str]) -> str {
    match r {
        Ok(n) if n > 100 => "large positive",
        Ok(n) if n > 0   => "small positive",
        Ok(0)            => "zero",
        Ok(n)            => "negative: " + str(n),
        Err(e)           => "error: " + e,
        // All cases covered -- no warning
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

61. Module System: Visibility

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles:

clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Details

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

```
// Pipeline with types at each stage
fn pipeline_example(nums: list[int]) -> map {
    let analysis = nums
        |> filter(fn(x) { x > 0 })                    // list[int]
        |> map(fn(x: int) -> float { sqrt(x as float) }) // list[float]
        |> take(10)                                     // list[float]
        |> fold({ "sum": 0.0, "count": 0 }, fn(acc, x) {
            acc["sum"]   = acc["sum"] + x
            acc["count"] = acc["count"] + 1
            return acc
        })  // map
    return analysis
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

62. Module System: Circular Dependencies

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

Details

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

```
// Async function and await
async fn fetch_and_parse(url: str) -> Result[Data, str] {
    let response = await http.get(url)
    let text     = await response.text()
    match parse_json(text) {
        Ok(json) => Ok(Data.from_json(json)),
        Err(e)   => Err("Parse error: " + e),
    }
}

async fn main() {
    let tasks = URLS |> map(fn(url) { fetch_and_parse(url) })
    let results = await gather(tasks)   // concurrent execution
    results |> filter(fn(r) { r.is_ok() }) |> for_each(println)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

63. Async/Await: Task Model

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in

two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Details

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

```
// Ownership and borrowing
fn process(data: list[int]) -> int {
    let total = sum(data)     // borrows data (immutable)
    let mean  = total / len(data)  // data still accessible
    return mean
}
// data is NOT moved -- process only borrows it

fn move_example() {
    let v = [1, 2, 3]
    let w = v               // v is MOVED into w
    // println(v)           // ERROR: v was moved
    println(w)              // OK
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

64. Async/Await: The Event Loop

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Details

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

```
// Module structure
module geometry

import math

pub struct Point {
    x: float,
    y: float,
}

pub fn distance(a: Point, b: Point) -> float {
```

```
    let dx = b.x - a.x
    let dy = b.y - a.y
    return math.sqrt(dx*dx + dy*dy)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

65. Foreign Function Interface (FFI)

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T:

Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Details

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

```
// Enum with data variants
pub enum Shape {
    Circle { center: Point, radius: float },
    Rect   { origin: Point, width: float, height: float },
    Triangle { a: Point, b: Point, c: Point },
}

pub fn area(s: Shape) -> float {
    match s {
        Shape.Circle { radius: r, .. }       => math.PI * r * r,

        Shape.Rect   { width: w, height: h, .. } => w * h,
        Shape.Triangle { a, b, c }           => triangle_area(a, b, c),
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

66. Macros

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Details

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

```
// Generic function with trait bound
trait Summable {
    fn zero() -> Self
    fn add(self, other: Self) -> Self

}

fn sum_all[T: Summable](items: list[T]) -> T {
    return items |> reduce(T.zero(), fn(acc, x) { acc.add(x) })
}

impl Summable for int {
    fn zero() -> int { return 0 }
    fn add(self, other: int) -> int { return self + other }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

67. Standard Library: Overview

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

Details

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async'

module provides async/await and the task scheduler.

```
// Pattern matching -- exhaustiveness
fn describe_result(r: Result[int, str]) -> str {
    match r {
        Ok(n) if n > 100 => "large positive",
        Ok(n) if n > 0   => "small positive",
        Ok(0)            => "zero",
        Ok(n)            => "negative: " + str(n),
        Err(e)           => "error: " + e,
        // All cases covered -- no warning
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

68. Standard Library: io Module

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

Details

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM

supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

```
// Pipeline with types at each stage
fn pipeline_example(nums: list[int]) -> map {
    let analysis = nums
        |> filter(fn(x) { x > 0 })                    // list[int]
        |> map(fn(x: int) -> float { sqrt(x as float) }) // list[float]
        |> take(10)                                     // list[float]
        |> fold({ "sum": 0.0, "count": 0 }, fn(acc, x) {
            acc["sum"]   = acc["sum"] + x
            acc["count"] = acc["count"] + 1
            return acc
        })  // map
    return analysis
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

69. Standard Library: net Module

Generics allow functions and types to be parameterized over types. A generic function 'fn id[T](x: T) -> T { return x }' works for any type T. A generic struct 'struct Pair[A, B] { first: A, second: B }' holds two values of potentially different types. Generic parameters can be constrained with trait bounds. Lateralus uses monomorphization: for each distinct set of type arguments, the compiler generates a specialized copy of the generic function or type. This produces efficient code (no boxing, no dynamic dispatch for generics) at the cost of potentially larger binary sizes.

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

Details

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

```
// Async function and await
async fn fetch_and_parse(url: str) -> Result[Data, str] {
    let response = await http.get(url)
    let text     = await response.text()
    match parse_json(text) {
        Ok(json) => Ok(Data.from_json(json)),
        Err(e)   => Err("Parse error: " + e),
    }
}

async fn main() {
    let tasks = URLS |> map(fn(url) { fetch_and_parse(url) })
    let results = await gather(tasks)   // concurrent execution
    results |> filter(fn(r) { r.is_ok() }) |> for_each(println)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

70. Standard Library: crypto Module

The standard library modules provide the building blocks of Lateralus programs. The 'collections' module provides list[T], map[K,V], set[T], deque[T], and heap[T]. The 'strings' module provides manipulation functions: split, join, trim, replace, starts_with, ends_with, contains, format. The 'io' module provides file I/O and console I/O. The 'net' module provides TCP/UDP networking. The 'math' module provides arithmetic functions, trigonometry, and numerical algorithms. The 'crypto' module provides cryptographic primitives (AES-GCM, ChaCha20-Poly1305, SHA-256, Argon2). The 'async' module provides async/await and the task scheduler.

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

Details

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

```
// Ownership and borrowing
fn process(data: list[int]) -> int {
    let total = sum(data)     // borrows data (immutable)
    let mean  = total / len(data)  // data still accessible
    return mean
}
// data is NOT moved -- process only borrows it
```

```
fn move_example() {
    let v = [1, 2, 3]
    let w = v               // v is MOVED into w
    // println(v)           // ERROR: v was moved
    println(w)              // OK
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

71. Standard Library: math Module

The Lateralus execution model supports both compiled (static) and interpreted (VM) execution. In compiled mode, the compiler generates Python, C, RISC-V, or x86-64 assembly, which is then passed to the host toolchain for final compilation. In VM mode, the compiler generates LBC bytecode, which is executed by the Lateralus VM. The VM supports JIT compilation: hot functions are detected by a sampling profiler and recompiled to machine code at runtime. The async runtime is built on a work-stealing task queue: async functions return immediately, scheduling a continuation to be executed when the awaited value is ready.

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The

'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

Details

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

```
// Module structure
module geometry

import math

pub struct Point {
    x: float,
    y: float,
}

pub fn distance(a: Point, b: Point) -> float {
    let dx = b.x - a.x
    let dy = b.y - a.y
    return math.sqrt(dx*dx + dy*dy)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

72. Standard Library: strings Module

Modules in Lateralus define the unit of compilation and the unit of capability declaration. Each source file is a module. A module begins with an optional 'module ModuleName' declaration followed by import statements and definitions. The 'pub' visibility modifier makes a definition accessible from other modules. By default, all definitions are private to the module. Re-exports allow a module to make imported names part of its public interface. The module system supports

circular imports through interface files (.lti): the compiler processes interface files before processing source files, breaking cycles by giving each module a declared interface before its implementation is compiled.

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Details

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if

expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

```
// Enum with data variants
pub enum Shape {
    Circle { center: Point, radius: float },
    Rect   { origin: Point, width: float, height: float },
    Triangle { a: Point, b: Point, c: Point },
}

pub fn area(s: Shape) -> float {
    match s {
        Shape.Circle { radius: r, .. }       => math.PI * r * r,
        Shape.Rect   { width: w, height: h, .. } => w * h,
        Shape.Triangle { a, b, c }           => triangle_area(a, b, c),
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

73. Standard Library: collections Module

The Lateralus programming language is a statically typed, expression-oriented language with first-class pipeline operators, algebraic data types, and a Hindley-Milner type inference engine. Its design is guided by three principles: clarity (code should read as naturally as prose), correctness (the type system should prevent entire classes of errors), and performance (zero-cost abstractions that compile to efficient machine code). This specification defines the normative syntax and semantics of the Lateralus language version 3.0, superseding previous specifications.

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a

sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Details

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

```
// Generic function with trait bound
trait Summable {
    fn zero() -> Self
    fn add(self, other: Self) -> Self
}

fn sum_all[T: Summable](items: list[T]) -> T {
    return items |> reduce(T.zero(), fn(acc, x) { acc.add(x) })
}

impl Summable for int {
    fn zero() -> int { return 0 }
    fn add(self, other: int) -> int { return self + other }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

74. Standard Library: testing Module

The Lateralus grammar is specified as an Extended Backus-Naur Form (EBNF) grammar. The top-level production is 'module', which consists of a module declaration, zero or more import statements, and a sequence of definitions. Definitions include function definitions, struct definitions, enum definitions, trait definitions, implementation blocks, type alias definitions, and constant declarations. The grammar is LL(1) for the statement level and uses Pratt parsing for expressions, providing unambiguous precedence climbing without explicit grammar rules for each precedence level.

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

Details

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

```
// Pattern matching -- exhaustiveness
fn describe_result(r: Result[int, str]) -> str {
    match r {
        Ok(n) if n > 100 => "large positive",
        Ok(n) if n > 0   => "small positive",
        Ok(0)            => "zero",
        Ok(n)            => "negative: " + str(n),
        Err(e)           => "error: " + e,
        // All cases covered -- no warning
    }
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

75. Compiler: Compilation Model

Expressions are the fundamental units of value computation in Lateralus. Every expression has a static type, inferred by the type checker. The expression forms are: literals (integer, floating-point, string, boolean, nil), identifiers, binary operations, unary operations, pipeline operations (|>), function calls, method calls, index expressions, struct construction, enum variant construction, match expressions, if expressions, block expressions, lambda expressions (fn(...) { ... }), and type cast expressions (value as Type). Function calls, match expressions, and blocks are all expressions that produce values.

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Details

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e),

Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

```
// Pipeline with types at each stage
fn pipeline_example(nums: list[int]) -> map {
    let analysis = nums
        |> filter(fn(x) { x > 0 })                    // list[int]
        |> map(fn(x: int) -> float { sqrt(x as float) }) // list[float]
        |> take(10)                                     // list[float]
        |> fold({ "sum": 0.0, "count": 0 }, fn(acc, x) {
            acc["sum"]   = acc["sum"] + x
            acc["count"] = acc["count"] + 1
            return acc
        })  // map
    return analysis
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

76. Compiler: Error Messages

Type annotations in Lateralus are optional but checked. When a programmer writes 'let x: int = 5', the compiler verifies that the inferred type of '5' is compatible with 'int'. When a function declares a return type 'fn foo() -> str', the compiler checks that all return paths produce a 'str'. Annotations are required in two situations: function parameters (to support separate compilation without full-program inference) and cases where the programmer wants to document intent explicitly. The type language includes: primitive types (int, float, str, bool, u8, u64), generic types (list[T], map[K,V], Option[T], Result[T,E]), function types (fn(A,B)->C), trait objects (dyn Trait), and user-defined types.

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

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Details

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

```
// Async function and await
async fn fetch_and_parse(url: str) -> Result[Data, str] {
    let response = await http.get(url)
    let text     = await response.text()
    match parse_json(text) {
        Ok(json) => Ok(Data.from_json(json)),
        Err(e)   => Err("Parse error: " + e),
    }
}

async fn main() {
    let tasks = URLS |> map(fn(url) { fetch_and_parse(url) })
    let results = await gather(tasks)   // concurrent execution
    results |> filter(fn(r) { r.is_ok() }) |> for_each(println)
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

impl declaration v1.0 implementation

match expression v1.0 pattern matching

|> operator v1.0 pipeline forward

async/await expression v1.3 concurrency

guard statement v1.4 early return guard

measure statement v1.4 performance profiling

emit statement v1.4 event tracing

from...import statement v1.5 selective import

77. Grammar Appendix: Complete EBNF

The ownership system tracks the lifetime of each value and ensures that values are not used after they have been freed or transferred. Every value has exactly one owner. When the owner goes out of scope, the value is freed. Values can be moved (transferring ownership) or borrowed (lending a reference without transferring ownership). Mutable borrows are exclusive: while a mutable borrow is active, no other borrows of the same value may exist. Immutable borrows are shared: multiple immutable borrows may coexist. The borrow checker, implemented as a dataflow analysis over the control flow graph, verifies that all borrows obey these rules at compile time.

Pattern matching in Lateralus provides exhaustive, structured destructuring. The 'match' expression takes a value and a sequence of arms, each consisting of a pattern and a result expression. Patterns can be: wildcard (_), variable binding (x), literal (42, "hello", true), enum variant (Some(x), Err(e), Color.Red), struct pattern ({x, y, z}), list pattern ([h, ...t]), tuple pattern ((a, b)), and guard pattern (pattern if condition). The compiler verifies that match expressions are exhaustive: every possible value of the matched type is covered by at least one arm. Missing patterns are compile errors with a suggestion of which cases are unhandled.

Traits define interfaces that types can implement. A trait declares a set of method signatures; implementing a trait requires providing a concrete implementation of each method. Traits support default method implementations, which can be overridden by implementations. A type implements a trait by providing an 'impl TraitName for TypeName' block. Trait bounds constrain type parameters: 'fn foo[T: Ord + Hash](x: T)' requires that 'T' implement both 'Ord' and 'Hash'. The compiler verifies trait implementations at the use site: if a function requires 'T: Ord', every call to that function with a concrete type 'T' is checked to ensure 'T' implements 'Ord'.

Details

Enums in Lateralus are algebraic data types -- tagged unions whose variants can carry data. An enum variant can hold any number of fields (unnamed tuple variants or named struct variants). The canonical pattern for nullable values is 'Option[T]' with variants 'Some(T)' and 'None'. The canonical pattern for fallible operations is 'Result[T,E]' with variants 'Ok(T)' and 'Err(E)'. The compiler's exhaustiveness checker ensures that all code paths handle all variants, preventing the class of bugs where new variants are added to an enum but some match expressions are not updated to handle them.

```
// Ownership and borrowing
fn process(data: list[int]) -> int {
    let total = sum(data)     // borrows data (immutable)
    let mean  = total / len(data)  // data still accessible
    return mean
}
// data is NOT moved -- process only borrows it
```

```
fn move_example() {
    let v = [1, 2, 3]
    let w = v               // v is MOVED into w

    // println(v)           // ERROR: v was moved
    println(w)              // OK
}
```

Keyword Kind Since Purpose

fn declaration v1.0 function definition

let statement v1.0 variable binding

mut modifier v1.0 mutable variable

pub modifier v1.0 public visibility

struct declaration v1.0 product type

enum declaration v1.0 sum type

trait declaration v1.0 interface

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Lateralus is an open-source, zero-dependency programming language. Project home: https://lateralus.dev. Source: github.com/bad-antics/lateralus-lang. Released under CC BY 4.0.